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NXP

X-MR-NAVQ95-V2

NOTES:

1. Interrupted lines coded with the same letter or letter combinations are electrically connected.
2. Device type number is for reference only. The number varies with the manufacturer.
3. Greyed out components indicate Do Not Populate (DNP) during assembly.
4. Special signal usage:
 - _B denotes an active-low signal.
 - _P and _N denote a differential signal.
5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.
6. All SILK display properties will be in this color
7. All schematic annotations will be in this color

Revision History

Rev.	Date	By	Description
X0-X13	2025-09-22	JvE (CTO SI)	Initial schematic development
A	2025-09-22	JvE (CTO SI)	Schematic release

A detailed changelog is available on page 26.

		System Innovation High Tech Campus 90 5656 AG Eindhoven, The Netherlands	
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© NXP SEMICONDUCTORS		Classification: <u>Company Internal/Proprietary</u>	
Designer: Jari van Ewijk	Drawing Title: X-MR-NAVQ95-V2		
Drawn by: Jari van Ewijk	Page Title: Cover		
Approved: Cristin Crutchei	Size: C	Document Number: SCH-95869 PDF: SPF-95869	Rev: A
Date: <u>Friday, September 26, 2025</u>		Sheet <u>1</u> of <u>26</u>	

DNP Components

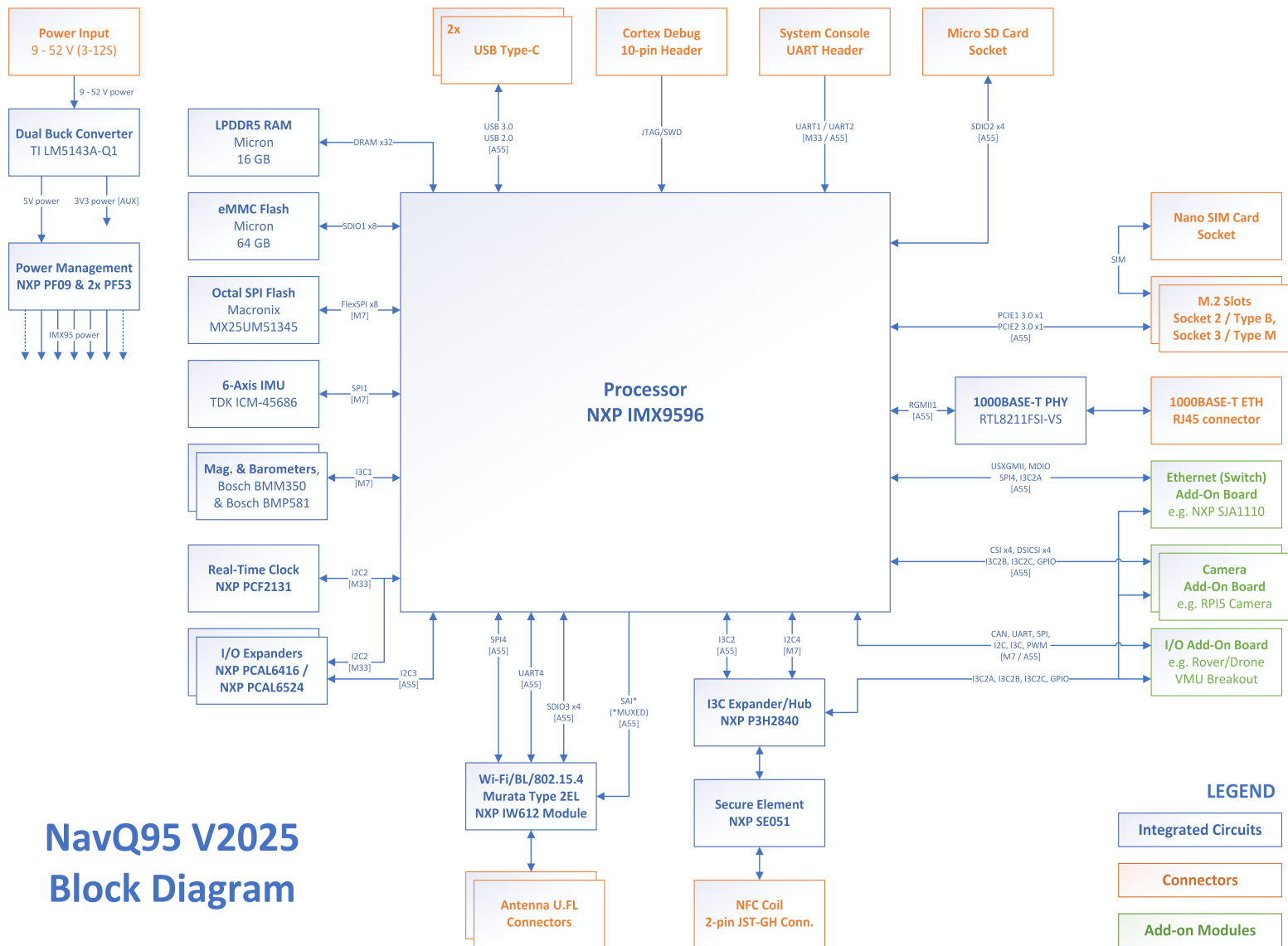
REF DES	ASSY OPT	PAGE NAME
R124,R126,R269	DNP	05 - HV Buck Converter
C23,C29,C92,R65,R68,R73,R74, R145,R152,R156,R229,R230,R257	DNP	06 - LV PMICs
R112,R123,R137,R138,R139,R254	DNP	09 - MPU I/O
C181,D14,R250	DNP	10 - MPU Misc.
R292,R295,R296,R297,R298	DNP	11 - eMMC Flash
R141	DNP	12 - Octal SPI Flash
C81,R288,R290	DNP	13 - MicroSD Card
R52,R56,R60,R75,R78	DNP	17 - RTC & Clock Buffer
R378,R379	DNP	18 - Sensors
C427,R158,R159,R162,R164, R165,R166,R169,R361,R362, R364,R365	DNP	19 - Tri-Radio
R315,R317	DNP	20 - Gigabit Ethernet
R212	DNP	21 - M.2 Key M
R113,R114,R115,R266,R267, R268,R289	DNP	22 - M.2 Key B
R176,R177,R178,R179,R180,R181	DNP	23 - I/O Expanders
R185	DNP	24 - Secure Element

Switch Default Positions

REF DES	SWITCH	PAGE NAME
SW2	2-4-5-7	14 - Buttons & Boot Mode



NavQ95 V2025 Block Diagram



LEGEND

Integrated Circuits

Connectors

Add-on Modules



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Page Title: **Block Diagram**

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Rev A

Molex Micro-Fit 2-pin power input
3S-12S BAT IN (6S recommended)
9-52V / 8A

LM5143 Buck Converter

PF09 / PF5301 / PF5302
VAON = 1.8V/10mA
VDD_SOC, PF5302 = 0.61-0.955V/15A
VDD_ARM, PF5301 = 0.61-1.05V/8A
SW2 = 0.8375/2.5A
SW3 = 1.8/2.5A
SW5 = 1.05-1.1V/2.5A
SW4 = 0.5-0.6V/2.5A
SW1 = 3.3V/3.5A
LDO2 = 1.8-3.3V/200mA
LDO1 = 3.3V/500mA
LDO3 = 0.65V/200mA
RSTB

IMX95
NVCC_BB5M_1V8 (2mA req.)
VDD_SOC (14A req.)
VDD_ARM (6.5A req.)
VDD_ANA_0P8 (1.5A req.)
VDD_ANAx_1P8 (300mA req.)
VDD_GPIO_1P8 (400mA req.)
VDD2_DDR (50mA req.)
VDDQ2_DDR (250mA req.)
VDD_GPIO_3P3 (400mA req.)
NVCC_SD2 (200mA req.)
SD_CARD (400mA req.)
POR_B

VDD1 = 1.8V/50mA
VDD2 = 1.05V/1.2 A
VDDQ = 0.5V/300mA
LPDDR5

VCC = 3.3V/200mA
VCCQ = 1.8V/250mA
eMMC

VDD = 3.3V
VIO = 1.8-3.3V
MicroSD

VCC = 1.8V/80mA
OctalSPI Flash
MX25UM51345GXD100

VDD = 3.3V/100nA
BACKUP BATTERY
RTC

AVDD18 = 1.8V/1A
AVDD33 = 3.3V/500mA
IW612

VDD = 3.3V/1A
RTL8211F5

VDD = 3.3V/50mA
156.25MHz
ETH REFCLK
e.g. 511DBA156M250BAG

VDD = 1.8V/50mA
100MHz
PCIe Clock Buffer
e.g. S153204-A01AGM

3.3V/50mA
1.8V/50mA
Sensors

3.3V/250mA
1.8V/250mA
I/O Expanders

RPI5-Compatible RTC Backup Battery
JST-SH 2-pin connector

VDD_SYS_SV / 4.5A
USB PD

VDD_SYS_SV / 1A
VDD_SYS_3V3 / 2A
I/O ADD-ON

VDD_SYS_SV / 1A
VDD_SYS_3V3 / 2A
ETH ADD-ON

VDD_SYS_SV / 1A
VDD_SYS_3V3 / 2A
CAM ADD-ON

VDD_SYS_3V3 / 2.5A
(5A peak for 100us)
M.2 Key B

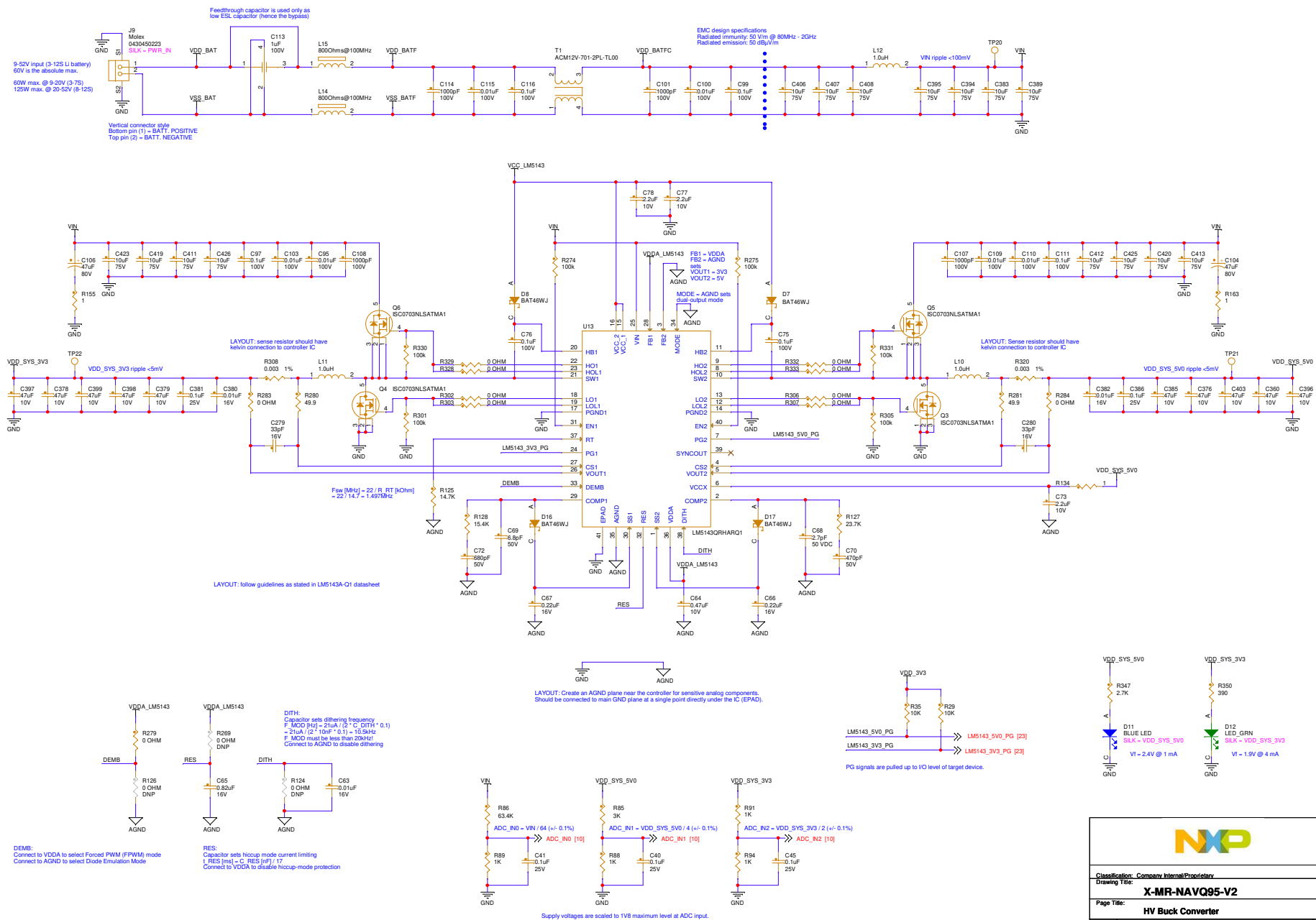
VDD_SYS_3V3 / 3.5A
(7A peak for 100us)
M.2 Key M

NavQ95 V2025 Power Tree



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HV Buck Converter



LV PMICs

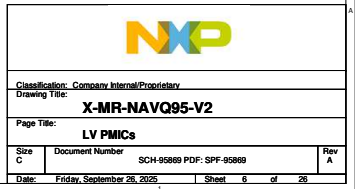
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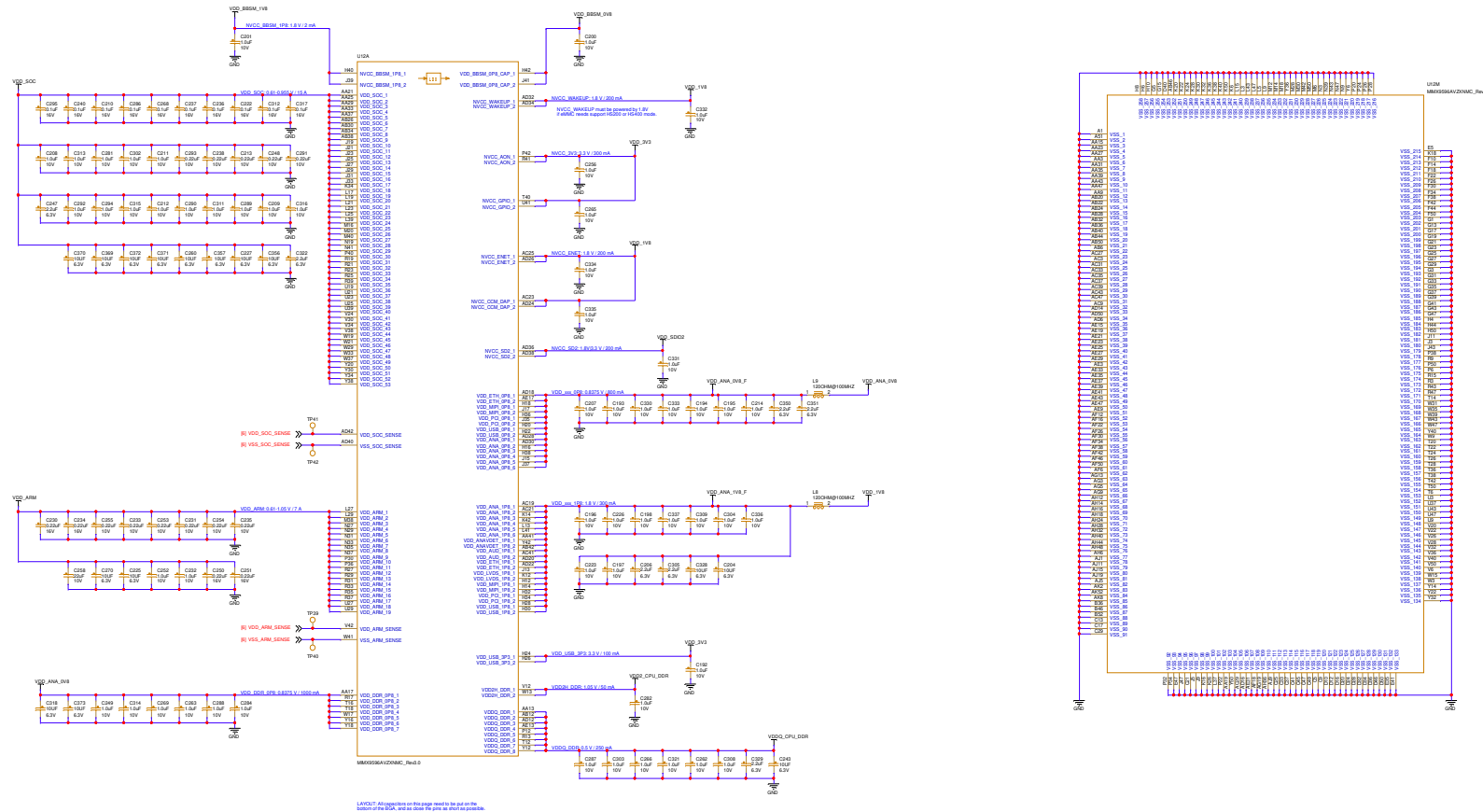
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Size C Document Number SCH-95899 PDF: SPF-95899

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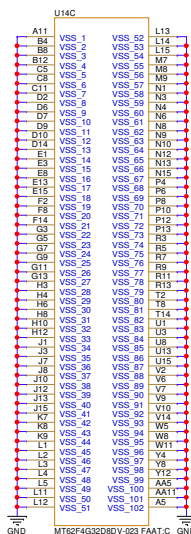
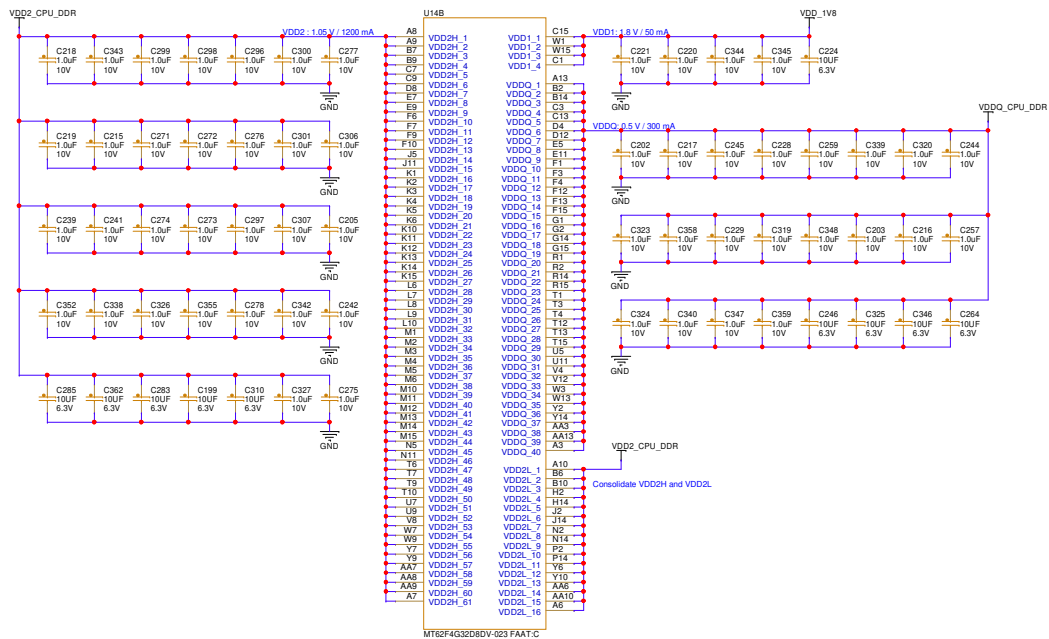
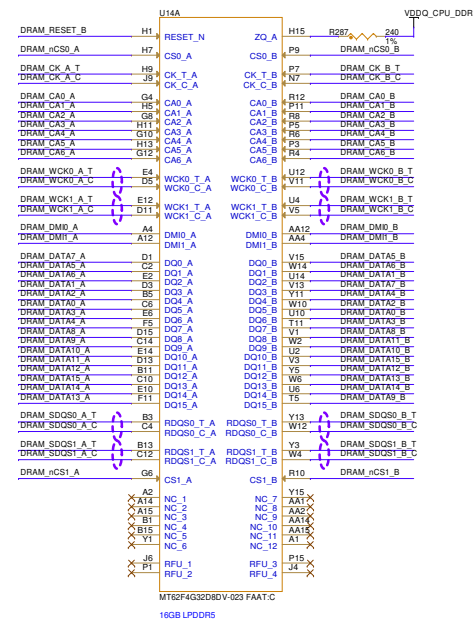
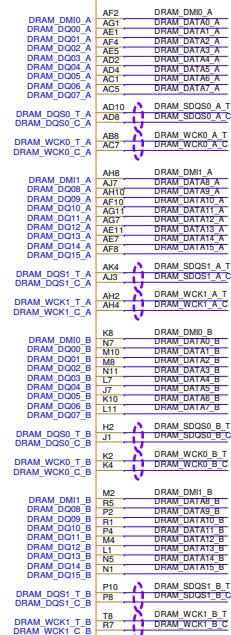
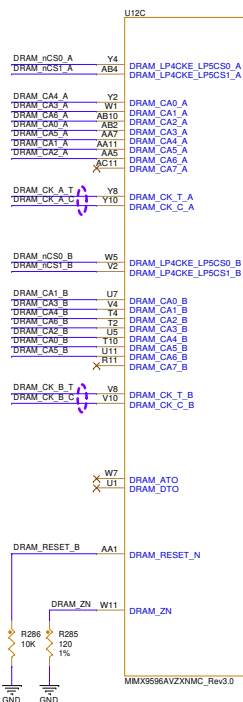


MPU Power



Note: net names are re-arranged (swizzled) to allow for optimized routing. CA net name follows memory device name. DQ net name follows CPU signal name.

DDR: 80 ohm



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X-MR-NAVQ95-V2

Page Title:

LPDDR5

Size

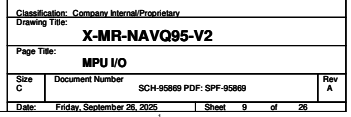
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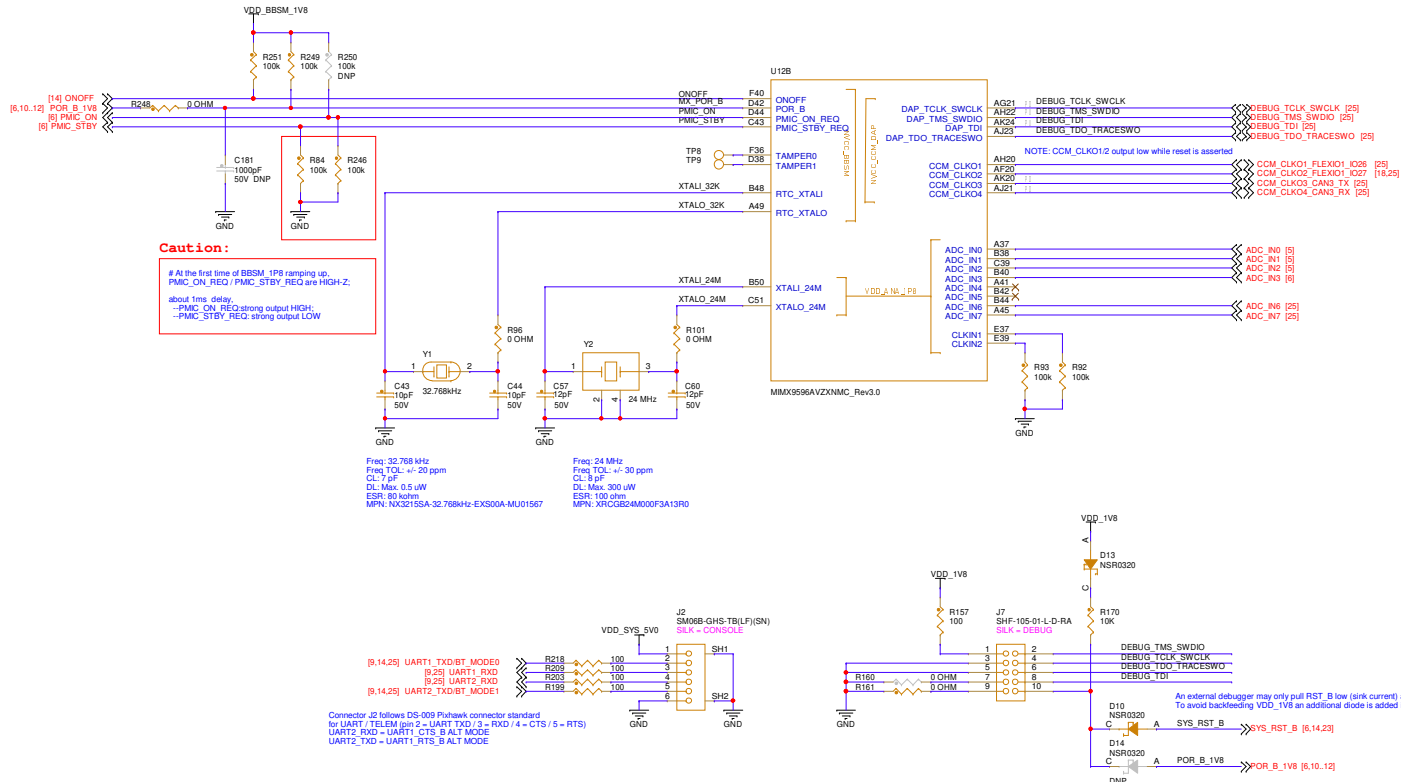
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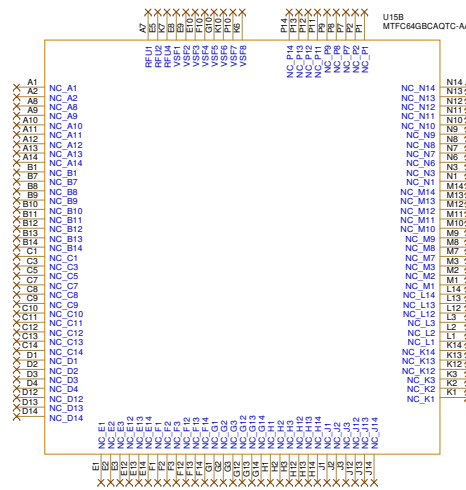
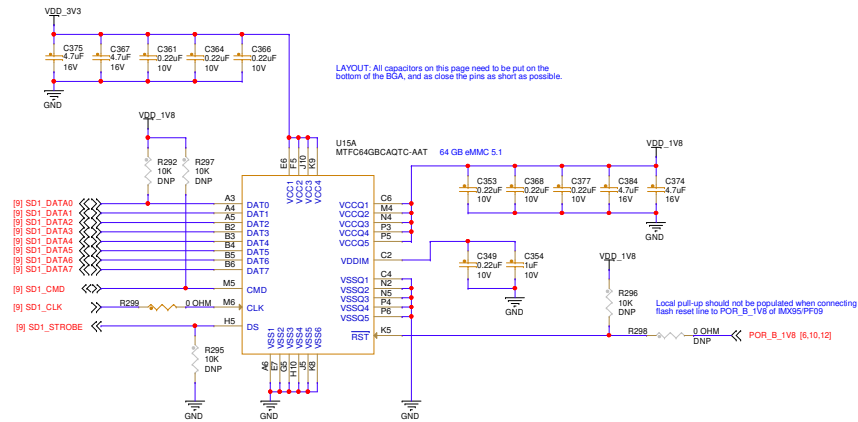
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MPU Misc.

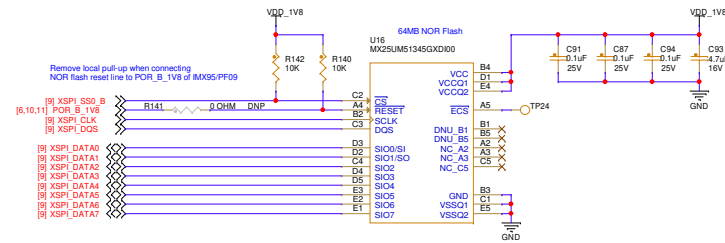


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Page Title: MPU Misc.			
Size: C	Document Number: SCH-95889 PDF: SPF-95889	Rev: A	
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eMMC Flash

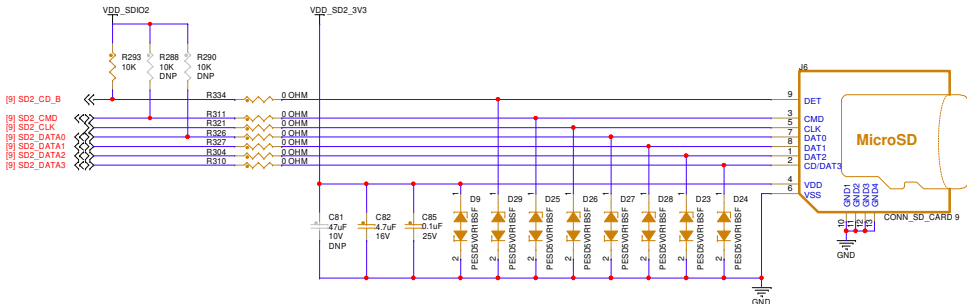


Octal SPI Flash



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Drawing Title: X-MR-NAVQ95-V2		
Page Title: Octal SPI Flash		
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MicroSD Card



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Drawing Title:

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Page Title:

MicroSD Card

Size	
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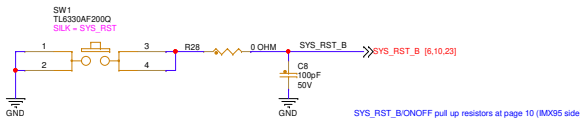
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Buttons & Boot Mode

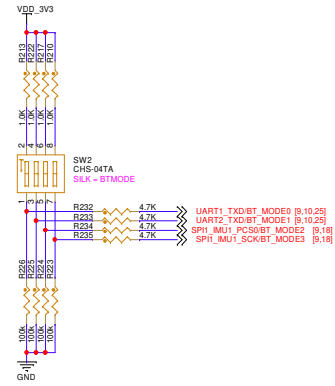
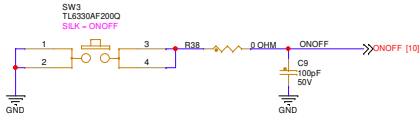
SYS RESET

Short Press: Reset



ONOFF

Long Press 5 seconds: Turn OFF
Short Press: Turn ON

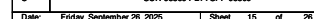


i.MX95 Boot Modes

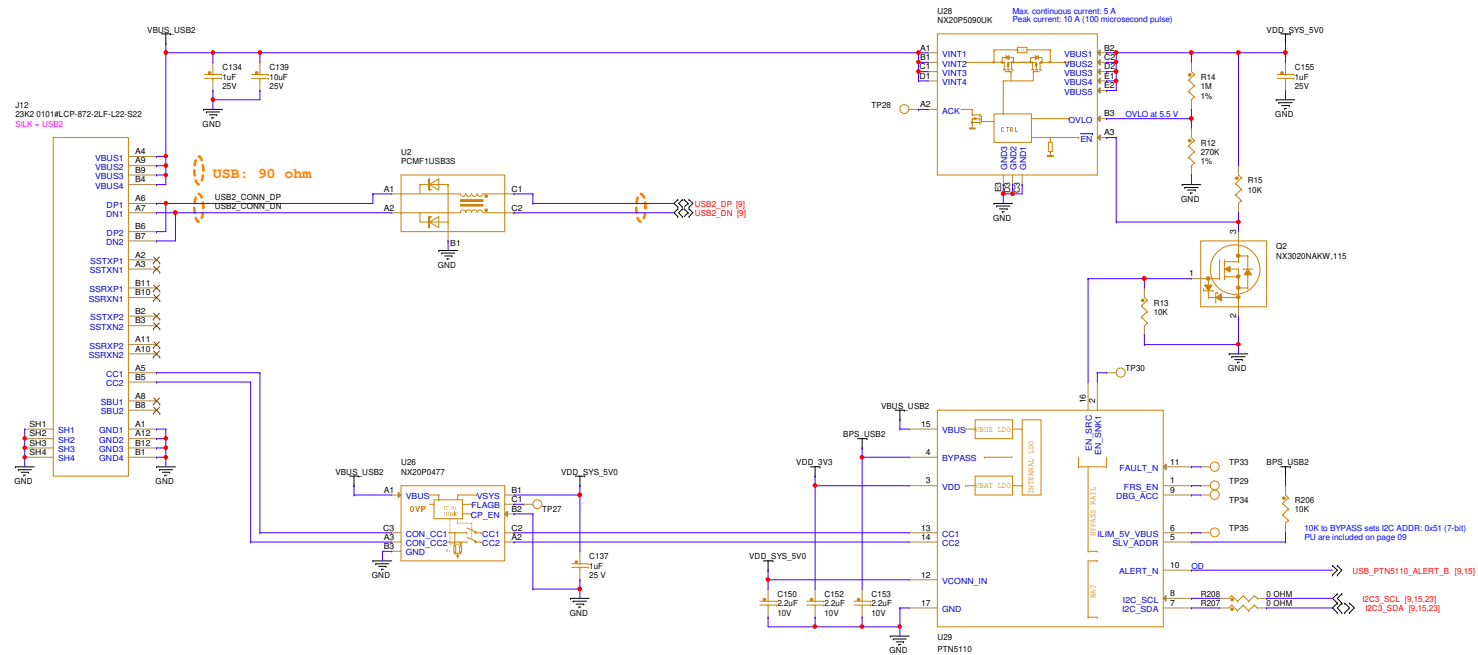
BOOT_MODE[3:0]	BOOT CORE	BOOT DEVICE	COMMENT
0001	Cortex-A55	Serial Downloader	USB1/2
0010	Cortex-A55	USDHC1 8-bit eMMC 5.1	
0011	Cortex-A55	USDHC2 4-bit SD3.0	
0100	Cortex-A55	FlexSPI Serial NOR	
0101	Cortex-A55	FlexSPI Serial NAND 2K page	
0110	Cortex-A55	FlexSPI Serial NAND 4K page	with SFDP (JESD-216) discoverable parameters
1001	Cortex-M33	Serial Downloader	
1010	Cortex-M33	USDHC1 8-bit eMMC 5.1	
1011	Cortex-M33	USDHC2 4-bit SD3.0	
1100	Cortex-M33	FlexSPI Serial NOR	
1101	Cortex-M33	FlexSPI Serial NAND 2K page	with SFDP (JESD-216) discoverable parameters
1110	Cortex-M33	FlexSPI Serial NAND 4K page	



A



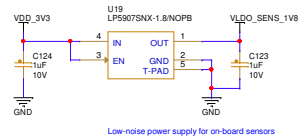
USB2 2.0 Type C



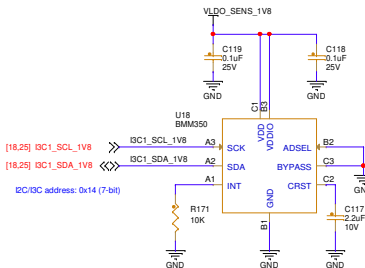
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Drawing Title: X-MR-NAVQ95-V2			
Page Title: USB2 2.0 Type C			
Size: C	Document Number: SCH-95889 PDF: SPF-95889	Rev: A	
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Sensors

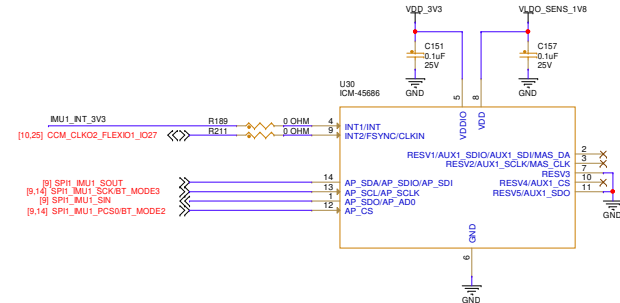
Low-Noise Regulator



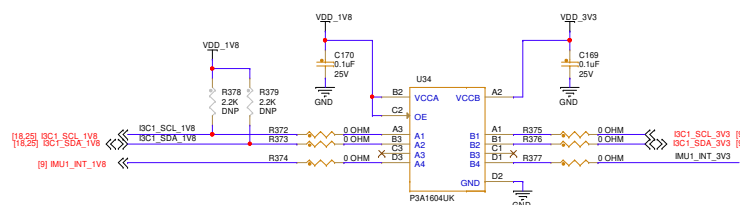
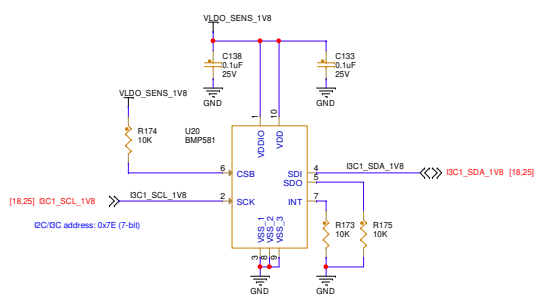
Magnetometer



6-Axis Accelerometer & Gyroscope



Barometer



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Wi-Fi 6 / Bluetooth 5.4 / IEEE 802.15.4



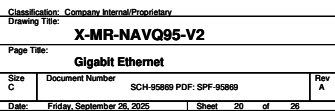
X-MR-NAVQ95-V2

Tri-Radio

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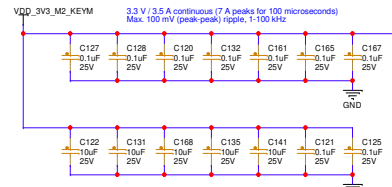
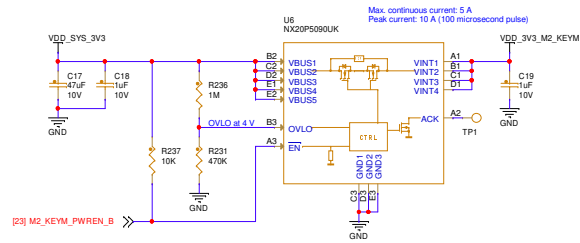
The schematic diagram illustrates the PCB layout for the X-MR-NAVQ95-V2, centered around the RTL8211FS-VS Ethernet controller. The controller is connected to various power and signal components:

- Power Management:** The controller is powered by AVDD10_RTL and AVDD33_RTL. It includes a 25 MHz oscillator (Y4) and several decoupling capacitors (C405, C414, C417, C409, C416, C418, C415, C424, C422, C428, C429, C430, C431, C432, C433, C434, C435, C436, C437, C438, C439, C440, C441, C442, C443, C444, C445, C446, C447, C448, C449, C450, C451, C452, C453, C454, C455, C456, C457, C458, C459, C460, C461, C462, C463, C464, C465, C466, C467, C468, C469, C470, C471, C472, C473, C474, C475, C476, C477, C478, C479, C480, C481, C482, C483, C484, C485, C486, C487, C488, C489, C490, C491, C492, C493, C494, C495, C496, C497, C498, C499, C500, C501, C502, C503, C504, C505, C506, C507, C508, C509, C510, C511, C512, C513, C514, C515, C516, C517, C518, C519, C520, C521, C522, C523, C524, C525, C526, C527, C528, C529, C530, C531, C532, C533, C534, C535, C536, C537, C538, C539, C540, C541, C542, C543, C544, C545, C546, C547, C548, C549, C550, C551, C552, C553, C554, C555, C556, C557, C558, C559, C560, C561, C562, C563, C564, C565, C566, C567, C568, C569, C570, C571, C572, C573, C574, C575, C576, C577, C578, C579, C580, C581, C582, C583, C584, C585, C586, C587, C588, C589, C590, C591, C592, C593, C594, C595, C596, C597, C598, C599, C600, C601, C602, C603, C604, C605, C606, C607, C608, C609, C610, C611, C612, C613, C614, C615, C616, C617, C618, C619, C620, C621, C622, C623, C624, C625, C626, C627, C628, C629, C630, C631, C632, C633, C634, C635, C636, C637, C638, C639, C640, C641, C642, C643, C644, C645, C646, C647, C648, C649, C650, C651, C652, C653, C654, C655, C656, C657, C658, C659, C660, C661, C662, C663, C664, C665, C666, C667, C668, C669, C670, C671, C672, C673, C674, C675, C676, C677, C678, C679, C680, C681, C682, C683, C684, C685, C686, C687, C688, C689, C690, C691, C692, C693, C694, C695, C696, C697, C698, C699, C700, C701, C702, C703, C704, C705, C706, C707, C708, C709, C710, C711, C712, C713, C714, C715, C716, C717, C718, C719, C720, C721, C722, C723, C724, C725, C726, C727, C728, C729, C730, C731, C732, C733, C734, C735, C736, C737, C738, C739, C740, C741, C742, C743, C744, C745, C746, C747, C748, C749, C750, C751, C752, C753, C754, C755, C756, C757, C758, C759, C760, C761, C762, C763, C764, C765, C766, C767, C768, C769, C770, C771, C772, C773, C774, C775, C776, C777, C778, C779, C780, C781, C782, C783, C784, C785, C786, C787, C788, C789, C790, C791, C792, C793, C794, C795, C796, C797, C798, C799, C800, C801, C802, C803, C804, C805, C806, C807, C808, C809, C810, C811, C812, C813, C814, C815, C816, C817, C818, C819, C820, C821, C822, C823, C824, C825, C826, C827, C828, C829, C830, C831, C832, C833, C834, C835, C836, C837, C838, C839, C840, C841, C842, C843, C844, C845, C846, C847, C848, C849, C850, C851, C852, C853, C854, C855, C856, C857, C858, C859, C860, C861, C862, C863, C864, C865, C866, C867, C868, C869, C870, C871, C872, C873, C874, C875, C876, C877, C878, C879, C880, C881, C882, C883, C884, C885, C886, C887, C888, C889, C890, C891, C892, C893, C894, C895, C896, C897, C898, C899, C900, C901, C902, C903, C904, C905, C906, C907, C908, C909, C910, C911, C912, C913, C914, C915, C916, C917, C918, C919, C920, C921, C922, C923, C924, C925, C926, C927, C928, C929, C930, C931, C932, C933, C934, C935, C936, C937, C938, C939, C940, C941, C942, C943, C944, C945, C946, C947, C948, C949, C950, C951, C952, C953, C954, C955, C956, C957, C958, C959, C960, C961, C962, C963, C964, C965, C966, C967, C968, C969, C970, C971, C972, C973, C974, C975, C976, C977, C978, C979, C980, C981, C982, C983, C984, C985, C986, C987, C988, C989, C990, C991, C992, C993, C994, C995, C996, C997, C998, C999, C1000, C1001, C1002, C1003, C1004, C1005, C1006, C1007, C1008, C1009, C1010, C1011, C1012, C1013, C1014, C1015, C1016, C1017, C1018, C1019, C1020, C1021, C1022, C1023, C1024, C1025, C1026, C1027, C1028, C1029, C1030, C1031, C1032, C1033, C1034, C1035, C1036, C1037, C1038, C1039, C1040, C1041, C1042, C1043, C1044, C1045, C1046, C1047, C1048, C1049, C1050, C1051, C1052, C1053, C1054, C1055, C1056, C1057, C1058, C1059, C1060, C1061, C1062, C1063, C1064, C1065, C1066, C1067, C1068, C1069, C1070, C1071, C1072, C1073, C1074, C1075, C1076, C1077, C1078, C1079, C1080, C1081, C1082, C1083, C1084, C1085, C1086, C1087, C1088, C1089, C1090, C1091, C1092, C1093, C1094, C1095, C1096, C1097, C1098, C1099, C1100, C1101, C1102, C1103, C1104, C1105, C1106, C1107, C1108, C1109, C1110, C1111, C1112, C1113, C1114, C1115, C1116, C1117, C1118, C1119, C1120, C1121, C1122, C1123, C1124, C1125, C1126, C1127, C1128, C1129, C1130, C1131, C1132, C1133, C1134, C1135, C1136, C1137, C1138, C1139, C1140, C1141, C1142, C1143, C1144, C1145, C1146, C1147, C1148, C1149, C1150, C1151, C1152, C1153, C1154, C1155, C1156, C1157, C1158, C1159, C1160, C1161, C1162, C1163, C1164, C1165, C1166, C1167, C1168, C1169, C1170, C1171, C1172, C1

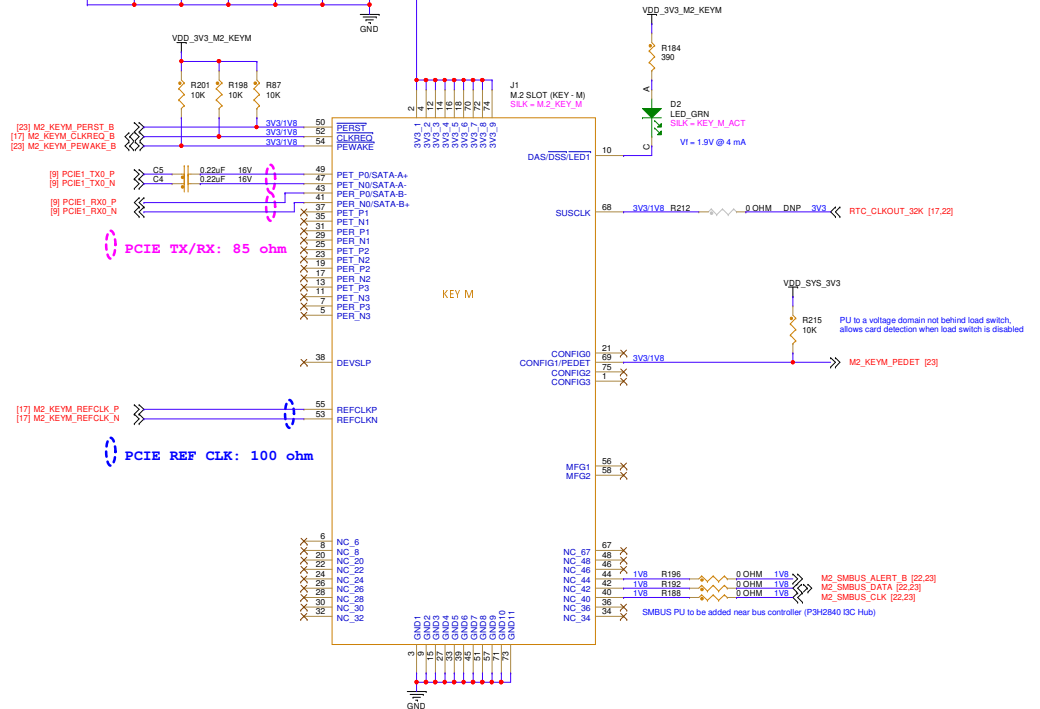


M.2 Key M

Load Switch



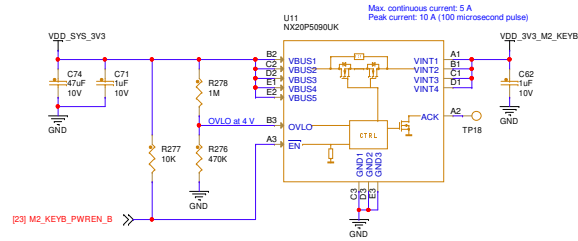
M.2 Socket 3 for Key M Cards



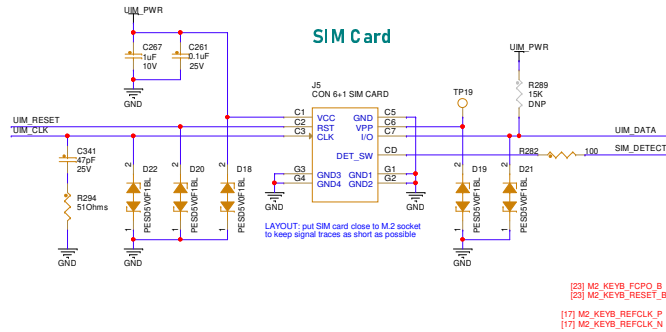
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M.2 Key B

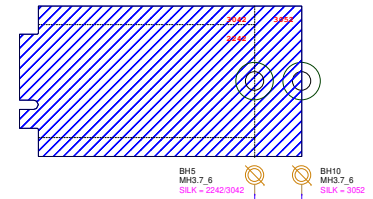
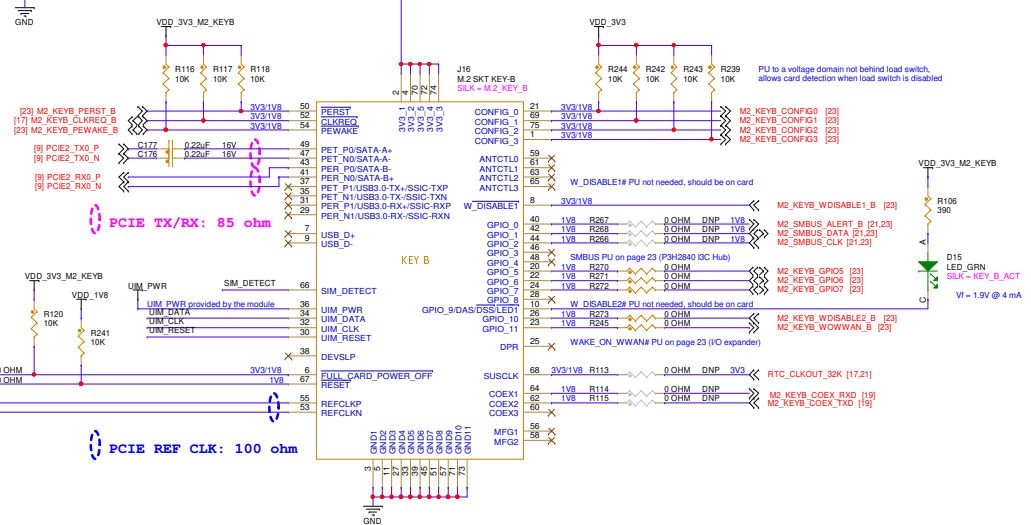
Load Switch



SIM Card

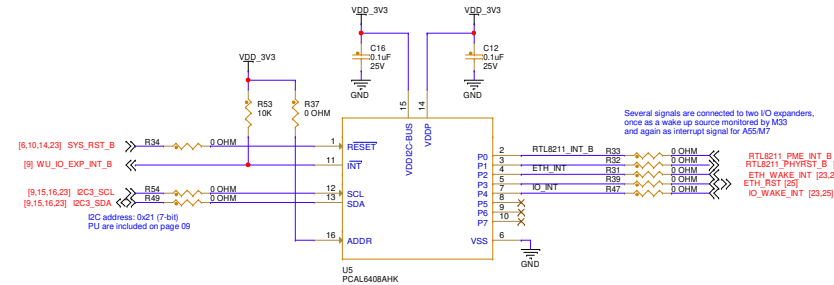
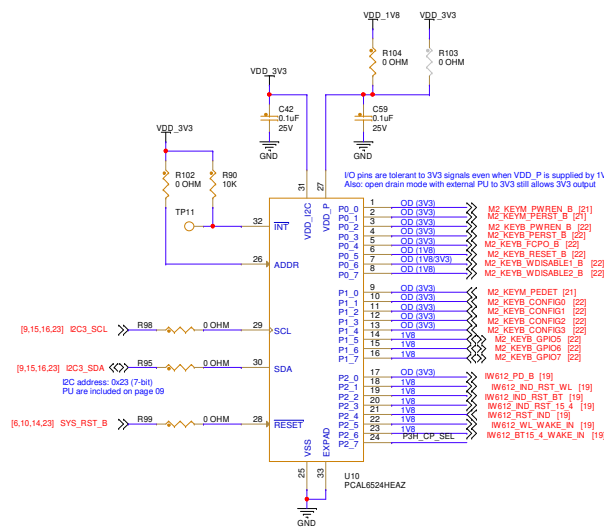
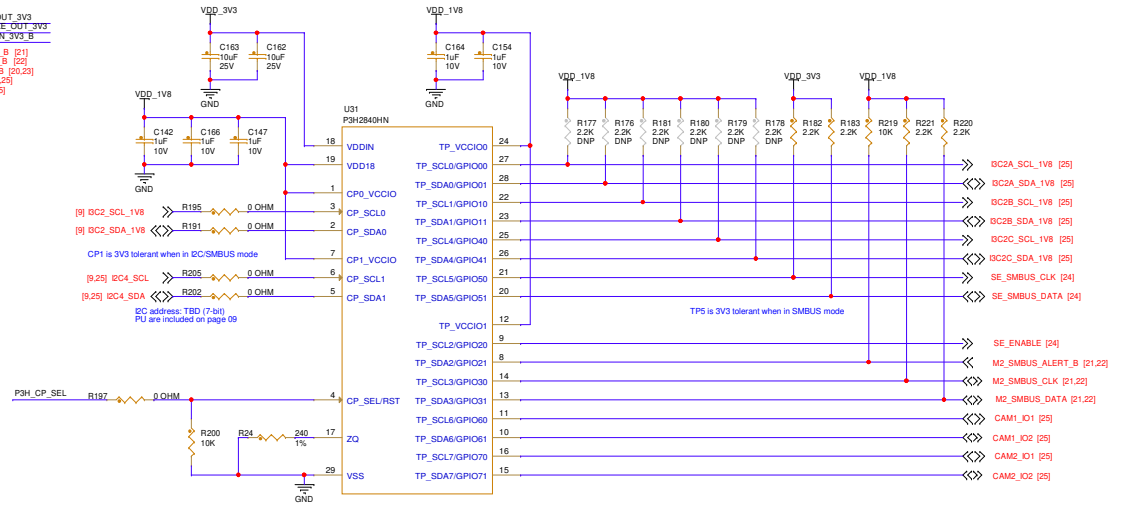
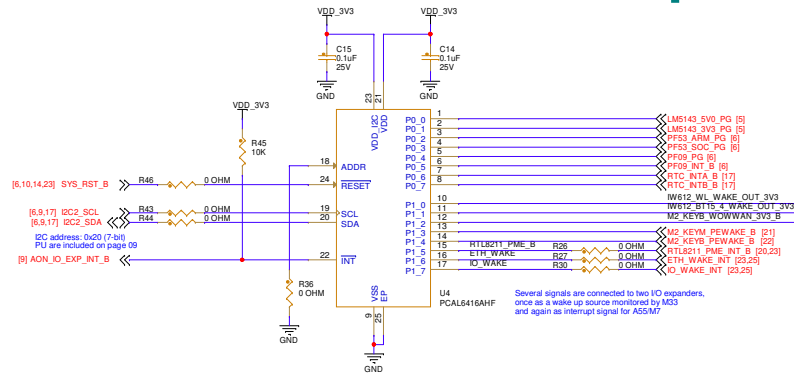


M.2 Socket 2 for Key B or B+M Cards



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I/O Expanders



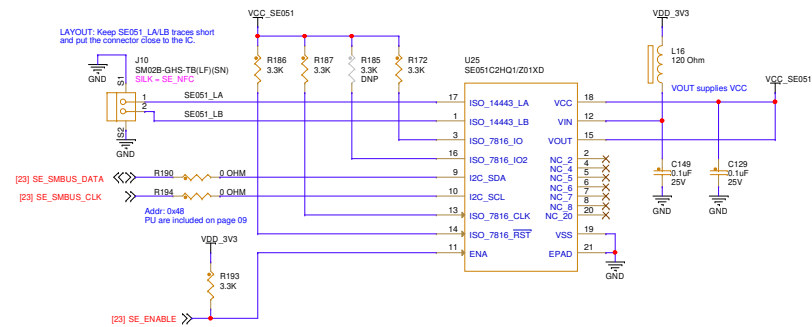
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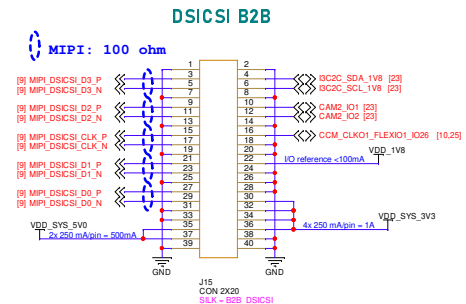
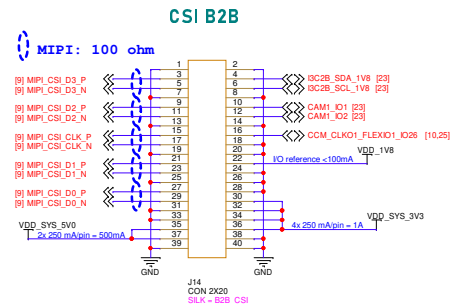
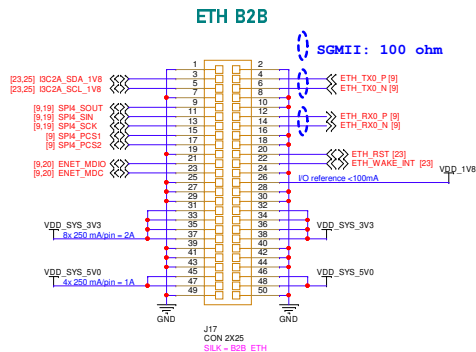
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Secure Element

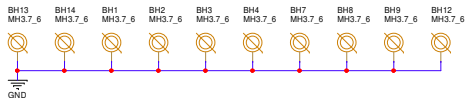
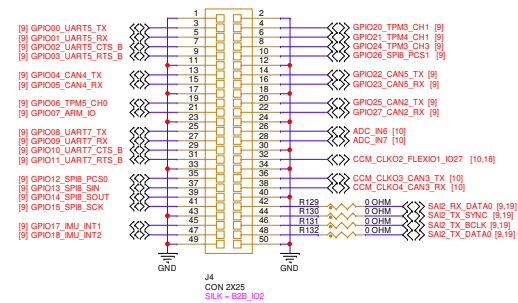
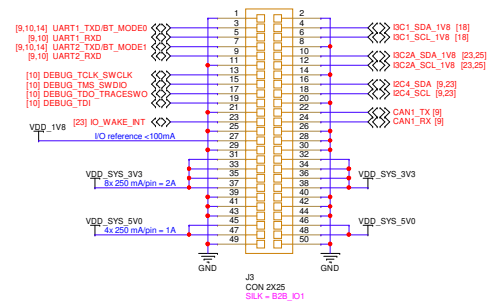


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Add-On B2B Interfaces



I/O B2B



Changelog

Rev.	Date	By	Description
X0	2025-07-04	JvE (CTO SI)	Initial version
X1	2025-07-10	JvE (CTO SI)	Release for A055 / preliminary BOM
X2	2025-07-21	JvE (CTO SI)	General cleanup of schematic. Pages 05-13 are ready.
X3	2025-07-22	JvE (CTO SI)	U1 (IMX95) part number updated. J22 replaced with different SIM card socket. Added JP1 header for console UART. Added SYSRST and ONOFF buttons.
X4	2025-07-24	JvE (CTO SI)	Visual updates / cleanup of pages 8 and 14-22 MPU I/O: Disconnect U1 C23 / USB1_DNU (previously USB1_ID) MPU I/O: Added pull up resistors for I2C3 and I2C4 USB1/2: Replace PTN5110 with a different variant USB1/2: NX20P5090UK swap VBUS and VINT connections RTC: Adjust R1246 for ML2020 RPI5 compatible battery Clock Buffer: Implement circuit ICM45686: Add 0R on INT & SYNC lines and connect to MPU Tri-Radio: Add in missing connections M.2 slots: Add in missing connections Replace P3H2840 symbol
X5	2025-07-25	JvE (CTO SI)	Ethernet: Replaced ix industrial connector with an RJ45 including integrated magnetics Ethernet: Remove some optional components around RTL8211 M.2 Key B: ESD protection added to SIM card M.2 Key B: Connections finished, PU and jumpers added M.2 Key B/M: Page / connection cleanup Tri-Radio: C683 added
X6	2025-07-31	JvE (CTO SI)	BOM Optimizations Tri-Radio: Add several 0 ohm resistors (can be used as probe point and can be used to add series resistance) Tri-Radio: Change VIO to 1V8, add levelshifter on SPI I/O Expanders: Add decoupling capacitors, required pull-ups, levelshifter for some signals.
X7	2025-08-08	JvE (CTO SI)	Change SGMII differential impedance to 100 ohm Replace R187 with 100k, matching the PF09 EVB Replace R44 with 10k Replace C401 with similar part used on the same page Replace J30 with SHF-105-01-L-D-RA (RA, TH) Replace JP1 with TSW-106-08-F-S-RA (RA, TH)
X8	2025-08-18	JvE (CTO SI)	ADC inputs: replace resistors with 0.1% precision parts Added D82/D83 ESD protection on RJ45 connector Swapped I2C SDA/SCL pins on PCF2131TFY Added U75 level shifter Replace JP1 header with connector J64
X9	2025-09-03	JvE (CTO SI)	Cover page updated. Table page added. Changelog page added. Annotations updated. Netnames updated.
X10	2025-09-04	JvE (CTO SI)	Schematic update after refdes renumbering.
X11	2025-09-08	JvE (CTO SI)	Update PCB footprints of C247, C260, C270
X12	2025-09-18	JvE (CTO SI)	Change R182/R183 reference voltage to 3V3 Add R371 PU on PTN5110 ALERT_B Add R378/R379 PU on U34 (DNP by default) Add R372-R377 series resistors on U34
X13	2025-09-22	JvE (CTO SI)	Adjust J2 pinout and change pin 1 power supply Swap SW2 pinout



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